

Can we find the eigenvalues of a matrix using Gaussian elimination? Why?

Answer:

No, in general, you cannot find the eigenvalues of a matrix using Gaussian elimination. The reason lies in the nature of Gaussian elimination and how it affects the matrix:

1. Gaussian Elimination Process:

Gaussian elimination involves applying a series of elementary row operations to transform a matrix into an upper triangular form or row echelon form. This process focuses on solving linear systems but does not preserve the eigenvalues of the original matrix.

2. Effect on Eigenvalues:

Eigenvalues are the roots of the characteristic polynomial. For two matrices **A**, **B** to have the same eigenvalues, if one can be converted into the other using similarity transformations (i.e. $\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}\mathbf{P}$) which involve both row and column operations). However, Gaussian elimination uses only row operations and alters the matrix in a way that does not preserve its eigenvalues.

3. Preservation of Linear Dependency:

While Gaussian elimination can be used to determine linear dependencies and the rank of a matrix, it is not suitable for finding eigenvalues because it does not maintain the structure necessary to preserve the characteristic polynomial.

4. Triangular Matrices Exception:

If a matrix is already triangular (upper or lower), then its eigenvalues can be read directly from its diagonal entries. However, getting a matrix into a triangular form via Gaussian elimination does not guarantee the eigenvalues are preserved.

Thus, the primary methods for finding eigenvalues include calculating the characteristic polynomial directly or using advanced algorithms (not in the course scope) that employ similarity transformations, such as the QR algorithm or diagonalization methods when applicable.